Accepted manuscript

This is the accepted manuscript of the article cited below. It has not been copyedited or typeset by the publisher, so its wording and formatting may differ from the published version. Please cite the published article:

Lynch, K., Chin, M., & Blazar, D. (2017). Relationships between observations of elementary mathematics instruction and student achievement: Exploring variability across districts. *American Journal of Education, 123*(4), 615–646.

https://doi.org/10.1086/692662

Relationships between Observations of Elementary Mathematics Instruction and Student Achievement: Exploring Variability Across Districts

Kathleen Lynch and Mark Chin

Harvard Graduate School of Education

David Blazar

University of Maryland College Park

Abstract

While much debate surrounding teacher quality has focused on students' standardized test scores, recent federal and state initiatives have emphasized the use of multiple measures to evaluate teacher quality, including classroom observations. In this study, we explore differences across school districts in the relationship between student achievement outcomes and the observed quality of teachers' instruction. Using data from 298 elementary mathematics teachers in five urban U.S. districts, we examine relationships between teachers' performance on the Mathematical Quality of Instruction (MQI) observation instrument and their students' scores on both state standardized and researcher-developed tests. We find that these relationships differ across school districts. We explore the extent to which differences in skills and expectations for students across tests may explain this variability. An improved understanding of the relationship between classroom observations and student tests may help districts to better support teachers in developing their instructional effectiveness.

Keywords: teacher evaluation, teacher effectiveness, classroom observations

Relationships between Observations of Elementary Mathematics Instruction and Student Achievement: Exploring Variability Across Districts

Spurred in part by federal initiatives such as Race to the Top and No Child Left Behind, school districts have accelerated development of new teacher evaluation systems that aim to improve student achievement and increase teachers' effectiveness (e.g. Firestone, 2014; Harris et al., 2014). While much of the policy debate regarding teacher quality has focused on student standardized assessments (Konstantopoulos, 2014), recent federal initiatives emphasize the role that classroom observations should play as a measure of teachers' professional practice (e.g., Coggshall et al., 2012).

However, to date, the evidence is mixed on the extent to which student achievement outcomes and classroom observation scores align to identify a common set of highly effective or ineffective teachers (e.g., Kane and Staiger, 2012; Kane et al., 2011). Correlations between classroom observation scores and student achievement outcomes generally vary across studies, from weak to moderate (e.g., Bell et al., 2012; Daley and Kim, 2010; Milanowski, 2004, 2011), although some research has found stronger relationships (Schacter and Thum, 2004).

Several theoretical and practical considerations underscore the need to understand why relationships between student achievement and teacher observation scores vary across contexts. On the one hand, perfect alignment between test scores and teacher observations is not expected; indeed, part of the rationale for including multiple measures in teacher evaluations is that these may provide broader information about instruction (Grossman et al., 2014). On the other hand, strong relationships between student test scores and teacher observation scores corroborate the logical expectation that better teaching will lead to increased student learning, and subsequently, improved test scores. Weak relationships challenge this theory, and may impede the use of

teacher evaluations in practice. Schools often use student achievement and teacher observation scores to inform important personnel decisions, such as merit pay or professional development (Podgursky and Springer, 2007). Misalignment between student test scores and teacher observation scores complicates the decision of which teachers to target for such interventions. For teachers, misalignment could potentially result in conflicting feedback on how to improve their practice. For example, if a teacher focused instruction relatively narrowly around basic skills, the teacher's students might show improvement on a basic skills-oriented state test, but the teacher might simultaneously receive low ratings on observational rubrics that emphasize more varied mathematics and instructional practices (Grossman et al., 2014; Polikoff, 2014).

In the current study, we take up the questions of whether differences in the relationships between teachers' observation scores and students' achievement outcomes exist across district contexts, and why these differences may exist. We explore an explanation for differences in relationships that until recently has received relatively little attention: the extent to which classroom observation instruments value teaching practices that are not well aligned to the skills expected of students on tests, resulting in *test-observation misalignment.*

To explore this issue, we use data from a sample of 298 elementary teachers teaching 6,780 students in five urban school districts, which were located in four different states. We examine the relationship between teachers' scores on the Mathematical Quality of Instruction (MQI), a classroom observation instrument developed to assess elementary mathematics teaching, and these teachers' students' achievement on two assessments: (1) a researcher-developed test, which was uniform across the five study districts; and (2) state standardized math tests. Instructional quality in all classrooms was measured using the MQI, meaning that teachers across districts were observed using the same metric.

Results from this study can provide guidance on features that may affect the alignment of observation instruments and student assessments used in new teacher evaluation systems.

Background and Research Context

Relationships Between Classroom Observations and Student Achievement Outcomes

There is a long history of research attempting to link characteristics of teachers and teaching to student achievement. Building from a broad tradition in the "education production function" literature of relating teacher characteristics (e.g., education, training, years of experience) to student outcomes, "process-product" studies beginning in the 1970s focused specifically on the relationship between teachers' classroom practices and student outcomes. Research found, for example, relationships between teachers' confidence, efficient use of class time, and group management to student achievement outcomes (for reviews, see e.g., Good and Brophy, 2007; Konstantopoulos, 2014). Research on opportunities to learn has found the amount and quality of students' exposure to new knowledge to be related to learning outcomes (e.g., Stevens, 1993). More recently, research using student surveys has found relationships between students' perceptions of teacher qualities such as care and challenge and student achievement (Kane and Cantrell, 2010). Critiques of these literatures have noted the largely correlational nature of findings, and the lack of focus on subject-specific instructional practices (Hill et al., 2005).

In recent years, researchers and practitioners have developed a range of new classroom observation instruments to assess the quality of teachers' instructional practices. The instruments were developed for a variety of purposes, ranging from the evaluation of early childhood instructional interactions and federally funded interventions (e.g. Gamse et al., 2008; Pianta et al., 2010) to instrument validation (e.g., Hill et al., 2008). From a policy standpoint, the

competition between states for federal Race to the Top grants, as well as the MET Project's research emphasis on rigor in teacher evaluation, further supported the reform and development of new teacher observation instruments (Hill et al., 2012). While classroom observation protocols were developed for several reasons, they shared the goal of providing observers across contexts standardized metrics to evaluate instruction.

Relationships between teachers' scores on observation instruments and student achievement outcomes vary across studies. With some exceptions (e.g. Schacter and Thum, 2004), these generally range from weak to moderate. Studies observing moderate relationships between these two measures found correlations/effect sizes in the range of 0.3 and 0.4 (Daley and Kim, 2010; Hill et al., 2010; Kane et al., 2011; Milanowski, 2004). However, several other studies have found relatively smaller relationships (e.g. Bell et al., 2012; Kane and Staiger, 2012; Milanowski, 2011). As an example of what was being related in these studies, in Daley and Kim (2010), the classroom observation tool assessed a mix of process-oriented (e.g., grouping students; lesson pacing) and content-oriented (e.g., presenting instructional content; problem-solving) elements, while student achievement was measured by state standardized tests.

Exploring Variability in the Relationships Between Teacher Observation and Student Achievement Scores

Extant research suggests a handful of possible explanations for varying relationships between teacher observation scores and student test scores. One category of explanations relates to the error with which such scores are measured. For example, researchers have found that teachers' value-added scores are inconsistent across years and sensitive to test timing (McCaffrey et al., 2009; Papay, 2011), yet gain precision when multiple years of data are utilized (Koedel and Betts, 2011). Observations scores also suffer from measurement error due to sources such as

variability in rater use of the instrument and the specific lessons observed (Hill et al, 2012; Bell et al, 2012; Kane and Staiger, 2012; Milanowski, 2011). Even before relating observation scores and student achievement, theory would indicate that measurement error in either or both of these measures would attenuate the strength of relationships (Spearman, 1904). Prior research has thus documented the impact of measurement error on teacher observation and student test scores; this suggests that it may also play a role in explaining cross-site differences in the relationships between these scores.

Beyond measurement error, however, relatively little research has explored which other factors might contribute to cross-district differences in the relationships between student test scores and teacher observation scores. In the current study, we explore the potential impact of misalignment between student tests and observation instruments in explaining these differences.

Classroom observation instruments may value certain teaching practices that are not well aligned to the skills expected of students on tests, resulting in *test-observation misalignment*. In this framework, different studies may return markedly different correlations between teacher observation scores and student achievement scores because of the differential sensitivity of the tests to different classroom practices. Polikoff (2014) has suggested that state tests differ in their instructional sensitivity, or the extent to which they reflect the content or quality of teachers' instruction. Low correlations between student test scores and classroom observation scores may reflect a discrepancy between the types of learning goals valued in classroom observations, versus the learning goals valued in standardized tests (Grossman et al., 2014; Polikoff, 2014). In particular, a core dimension along which standardized tests and classroom observations may differ is the cognitive demand of the learning activities they measure (Doyle, 1988). In Doyle's (1988) framework, low cognitive demand activities include memorizing facts and applying

formulas and procedures without attention to meaning. Stein and colleagues (1996) argue that classrooms dominated by these activities "do not provide the conditions necessary for the development of students' capacity to think and reason mathematically" (p. 457). By contrast, high cognitive demand activities generally involve comprehension, interpretation, or synthesis, include solving problems using multiple solution strategies, comparing representations, and explaining and justifying ideas (Doyle, 1988).

Previous research has found that state standardized tests often assess relatively low-cognitive demand skills, such as using procedures and applying formulas (e.g., Resnick et al., 2004; Webb, 1999). For example, in a study analyzing state assessments in mathematics and language arts, Resnick and colleagues (2004) found that state assessments items were generally inappropriately easy relative to the levels of cognitive demand indicated in state curriculum standards. In addition, state tests are frequently comprised mostly of multiple-choice items. While open-ended items do not automatically indicate a higher level of cognitive demand than multiple choice items, they are nonetheless theorized to provide the potential to pose higher levels of cognitive demand, because they cannot be solved by working backwards from a list of answer choices, and because open-ended writing may offer opportunities to assess distinct logical abilities (Bridgeman, 1992).

On the other hand, as discussed below, classroom observation instruments often measure the extent to which instruction facilitates high-cognitive demand activities, such as comparing, explaining, and justifying. This suggests that on observation measures, teachers may be rewarded for imbuing content with meaning or providing cognitively demanding activities; on the other hand, on some standardized tests, students may be asked primarily to demonstrate relatively low-level skills. In this example, the relationship between teachers' observation scores and their

students' achievement may be weak because these measures are poorly aligned. Indeed, this problem has received attention from policymakers; with the increased interest in more rigorous curriculum frameworks such as the Common Core State Standards for Mathematics (National Governors Association Center for Best Practices, 2010), this issue of misalignment between the content of what is taught and what is assessed has spurred work on the development of more conceptually oriented, rigorous assessments, designed to better assess higher-level competencies (Grossman et al., 2011).

To date, we have been able to identify only three prior studies that have explored differences in the types of skills expected of teachers and students in an observation instrument versus in student assessments, all using data from the MET project. MET researchers (Kane and Staiger, 2012; Grossman et al., 2014) observed that the relationships between teacher scores on observation instruments used in the MET study and student achievement were stronger when student achievement was measured using the Stanford 9 Open-Ended (SAT-9 OE) reading assessment than when measured using state standardized tests. Kane and Staiger (2012) speculated that this may have been due to differences between the two tests. They suggested that most state tests do not ask students to write about what they have read, a key skill that is both measured on the SAT-9 OE and a major focus of reading teachers' instruction.

Also using MET data, Grossman et al. (2014) examined how the relationship between teachers' scores on the Protocol for Language Arts Teaching Observations (PLATO) instrument and their value-added (VA) scores differed depending on whether student learning was assessed using state tests or the SAT-9 OE. They found that the relationship between teachers' PLATO scores and VA scores did differ depending on the student assessment used to construct the VA scores. Teachers' PLATO scores were more correlated with their VA scores when estimated

from the SAT-9 OE test than from state standardized tests. Grossman et al. suggested that higher correlations between students' SAT-9 OE performance and teachers' PLATO scores could be due to the fact that the SAT-9 OE was designed to capture students' argumentation skills and skill in developing ideas – skills also valued by the PLATO observation protocol – while they presumed that the state tests tended to emphasize lower-level comprehension skills and multiple-choice questions.

In addition, Polikoff (2014) examined MET data disaggregated by state. He found that the correlations between VA scores and teacher scores on several observation instruments varied substantially across the six MET states. Polikoff concluded that some states' tests were more correlated with teacher performance on certain classroom observation instruments than others; however, "the reasons for these differences are not well known, but they should be an intense focus of study in the coming years" (p. 301). Polikoff suggested several hypotheses, including that variability in correlations may be due to characteristics of the test items or content (such as low-level/procedural versus higher-level cognitive demand, or multiple choice versus open-response item format). We explore these hypotheses in the current study.

Theory of Instruction and Student Learning Underpinning the MQI

In this article, we use the MQI as an example of a mathematics-specific observation instrument emphasizing conceptually-oriented instruction, to illustrate the potential contrasts between the teaching and learning goals valued on this type of instrument versus those valued on standardized tests. Below, we describe the theory of instruction and student learning underpinning the MQI.

Underlying a theory of instruction is a vision of what constitutes high-quality student learning (Grossman et al., 2014). While there is a history of debate in mathematics education

regarding the relative importance of students' conceptual versus procedural learning (e.g., Schoenfeld, 2004), in recent years scholars have argued that procedural knowledge can be superficial as well as deep (Star, 2005; 2007), and influential policy documents have expressed the view that high-quality student learning encompasses both rich conceptual understanding as well as procedural fluency and skill development. Moreover, these are complementary and intertwined (e.g., Common Core; National Research Council's *Adding It Up* [2001]).

Prior research has examined which elements of instruction support the development of high-quality student learning in mathematics. In their influential article, Hiebert and Grouws (2007) reviewed the literature on the effects of classroom mathematics teaching on student learning in the areas of conceptual understanding, defined as 'mental connections among mathematical facts, procedures, and ideas,' and skill efficiency, defined as 'the accurate, smooth, and rapid execution of mathematical procedures' (p. 380). A limitation of this conceptualization is that it leaves out other important goals, such as the ability to apply procedural skills flexibly and strategically in novel situations (e.g., Star, 2005, 2007; Star et al., 2014); however, it nonetheless emphasizes two important and longstanding goals of school mathematics learning. Hiebert and Grouws identified two related sets of instructional interactions, which we refer to as *conceptually-oriented* and *skill efficiency-oriented*. Two key elements of conceptually-oriented instruction were found to support students' *conceptual learning*: (1) teaching attended explicitly to concepts, or connections among facts, procedures, and ideas; and (2) students expended effort to make sense of important mathematics (e.g., Boaler, 1998; Carpenter et al., 1989; Stein and Lane, 1996).

With regard to whether conceptually-oriented or skill efficiency-oriented instruction better supported students' *skill learning*, Hiebert and Grouws's findings were less conclusive. They

write that, “One place in which the complex nature of teaching and learning becomes apparent is in the effects of conceptually oriented teaching on skill learning. Many of the reports on the conceptual development of students also indicate that their skills increased at a level equal to or greater than students in the control groups … Apparently, it is not the case that only one set of teaching features facilitates skill learning and another set facilitates conceptual learning. In this case, two quite different kinds of features both seem to promote skill learning.” (pp. 390). The research suggested that skill development could also be bolstered under a quite different set of instructional conditions, ‘skill efficiency-oriented.’ *Skill efficiency-oriented* instruction involved rapid pace, short-answer targeted teacher questions, and students completing large numbers of practice problems (Hiebert and Grouws, 2007).

This lack of certainty as to whether conceptually-oriented teaching is more effective or equally effective as skill efficiency-oriented teaching at promoting skill learning makes the issue of alignment more complicated. Consider the (common) scenario where teacher evaluations are based predominantly on classroom observation scores and relatively rote standardized tests. If conceptual teaching is in fact *more effective* than skill efficiency teaching at improving skills, then teachers using conceptual teaching should receive scores on both classroom observations and student tests that are high relative to those using skill efficiency teaching (although perhaps not as high as they should be, as noted below). However, if conceptually-oriented teaching is actually *equally effective* as skill efficiency-oriented teaching at improving skills, then a teacher using conceptually-oriented teaching could potentially receive a conflicting evaluation report, with high classroom observation scores but average test score gains similar to those of her peers using skill efficiency-oriented teaching methods. In either case, if we hypothesize that conceptually-oriented teaching methods also have greater affordances for students’ conceptual

learning than do skill efficiency-oriented teaching methods, then conceptually-oriented teachers' students' test score gains on skills-oriented tests may underestimate the impacts these teachers have on student learning, because conceptual outcomes are not measured on the tests. Related to this, a second potential misalignment issue under this scenario is that skills-oriented test content is misaligned to standards advanced by NCTM and the Common Core State Standards initiative (National Governors Association Center for Best Practices, 2010; NCTM, 2000). See Figure 1 for a graphical depiction of the potential areas of test-observation misalignment under these hypotheses.

In contrast with standardized tests, which often measure low-level skills (Resnick et al., 2004), classroom observation instruments are often grounded in a theory of instruction aligned with conceptually-oriented instruction, emphasizing the quality of teacher-student interactions and the development of content understanding via engagement in cognitively demanding activities (Grossman et al., 2014; Seidel and Shavelson, 2007). 'Content-generic' instruments are designed to assess instructional interactions across subject areas (e.g., Framework for Teaching [Danielson, 2011]; CLASS [Pianta et al., 2010]), while 'content-specific' instruments assess subject-specific instructional quality (Charalambous et al., 2014). Examples of mathematics content-specific instruments include the Instructional Quality Assessment (IQA) (Boston, 2012), which is rooted in research on challenging mathematical tasks and maintaining cognitive demand (e.g., Stein and Lane, 1996). The Reformed Teaching Observation Protocol (RTOP) (Sawada et al., 2002) was developed to measure instructional alignment with mathematics reforms which emphasized a problem-solving approach. (e.g., NCTM, 1995, 2000).

The MQI grew out of research on the importance of teachers' mathematical knowledge for teaching and precision in mathematical discussions and explanations. For the MQI, high-quality

student learning involves reasoning and making meaning about the mathematical content, with the goal of developing conceptual understanding, adaptive reasoning, and procedural fluency. Note that we do not argue that the MQI instrument reflects all aspects of strong teaching as reflected in the mathematics education literature. Indeed, there are many alternative perspectives beyond those reflected in the MQI which could be used to evaluate the quality of teachers' instruction, such as gauging student discourse (e.g., O'Connor, 1999) and charting in detail the cognitive demand of unfolding tasks (e.g., Stein and Lane, 1996). It was not possible for us to measure all of the relevant constructs, and important ones are certainly left out. Nonetheless, we argue that the MQI is appropriate for our analyses because it measures several constructs which the mathematics education literature has identified as related to high-quality instruction in mathematics, as noted below.

Specifically, the MQI is grounded in a theory of *ambitious instruction* which emphasizes the importance of three core areas of instructional interactions: (1) *teacher-students*, (2) *teacher-content*, and (3) *students-content* (see Figure 2) (Hill et al., 2008). 'Ambitious instruction' refers to instructional practice that is intellectually demanding and attentive to students' work (Cohen, 2011). See Table 1 for a description of the MQI dimensions and items. In interactions area (1), *teacher-students*, the MQI captures teachers' skill in responding to students' mathematically substantive productions, including utterances, written work, and mathematical errors. For example, teaching practices in this MQI domain include the skill with which teachers identify mathematical insight in specific student questions, comments, or work; and build instruction on student ideas or methods. The elements in this domain are grounded in research on the importance of teachers' responses to students' mathematical questions and ideas (e.g., Borko, et al., 1992; Cohen, 1990; Stein et al., 1990), for example, Stein, Engle, Smith, and Hughes's

(2008) argument for the importance of teachers helping students to draw out the connections between the mathematical ideas underpinning the strategies they use.

Interactions area (2), *teacher-content*, captures the depth and richness of the mathematics offered to students, where rich mathematics focus on the meaning of facts, procedures, and practices. The elements in this domain are drawn from research in the mathematics education literature which highlights the affordances of linking and connecting mathematical representations, ideas, and procedures; considering multiple solution methods; and giving mathematical meaning to ideas or solution methods (e.g., Fennema and Franke, 1992; Lloyd and Wilson, 1998; Ma, 1999; Stigler and Hiebert, 1999). For example, one teaching practice evaluated in this dimension is the extent to which teachers draw explicit links among mathematical representations, ideas, and procedures, using utterances, written work, and gestures (e.g., Richland, 2015). Another practice measured in this domain is whether the teacher facilitates students to consider and compare multiple solution strategies or procedures for a single problem (e.g., Star and Rittle-Johnson, 2009).

Interactions area (3), *students-content*, captures students' participation in cognitively activating work. For example, practices captured in this dimension include the extent to which students provide explanations (e.g., Henningsen and Stein, 1997; Stein et al., 1996), pose mathematically motivated questions or advance mathematical claims or counterclaims (Yackel and Cobb, 1996); draw connections among different representations, concepts, or solution methods (Star et al., 2015); and engage in reasoning and other cognitively demanding activities, such as identifying and explaining patterns.

In work drawing on a subset of the data used in this analysis, Blazar (2015) found that the *ambitious instruction* measure from the MQI did predict gains in students' mathematics

achievement on a researcher-developed test when pooling across teachers in multiple districts. A one standard deviation increase in teachers' *ambitious instruction* scores was associated with roughly 0.10 standard deviation increase in students' mathematics achievement. That work did not explore differential relationships between *ambitious instruction* and student achievement across different standardized assessments. For a detailed discussion of the MQI dimensions and supporting research, also see Blazar et al, forthcoming and Hill et al., 2008.

Contribution of the Current Study

In summary, our work builds on prior studies in the following ways. First, our data allow us to investigate relationships between teacher observation scores, as measured by the MQI instrument, and student achievement across districts where students take different state standardized math tests, but also a researcher-developed math test common to all students. Second, we conduct a formal coding analysis of tests used in the study, in order to explore more systematically the hypothesis that tests' cognitive demand and item and format characteristics are related to test-observation misalignment.

Because we explore these possible explanations in a single study, we compile more evidence regarding certain explanations for varying relationships between student assessments and teacher observations than currently exists in the literature. However, our study design does not permit us to disentangle the specific impact of test-observation misalignment. In addition to the variables we have measured here, educational outcomes necessarily reflect the complex interplay of students and teachers, resources, and the broader social, cultural, and historical context (Cohen et al., 2003). We do not address the latter contextual issues in the current study, but rather focus specifically on the issue of test-observation misalignment. As such, the current study comprises a preliminary examination of an important, yet limited set of constructs which may bear on

relationships between teacher observation scores and student achievement. We return to this issue in the discussion.

Research Questions

We ask the following research questions: Does teacher performance on observation instruments predict student achievement equally well across different district contexts? To the extent that relationships between teacher observation scores and student achievement scores vary across districts, we explore possible factors that may contribute to this variability.

Method

Sample

Data for this study come from two research projects which spanned the 2010-2011 and 2011-2012 school years, and focused on fourth- and fifth-grade mathematics teachers. The first study is a large-scale project examining instructional quality in four anonymous districts (henceforth, districts 1 through 4) from three states. The second is a randomized controlled trial of a mathematics professional development program in one anonymous district (henceforth, district 5); this study collected similar data on teachers and students as the first project. In the first project, schools were recruited based on district referrals and size (with a minimum of two teachers in each school in each sampled grade). Of eligible teachers in these schools, roughly 55% agreed to participate. In the second study, we only include the treatment teachers, as MQI data were not collected for the control group teachers during these school years. We restrict this sample to the teachers for whom we have observation data, state test score data, and researcher-developed assessment score data, for a total sample of 298 teachers and 6,780 students. From

these 298 teachers, we collected 1,560 videotaped lessons of mathematics instruction across both school years.

In Table 2, we present descriptive statistics for the sample teachers and their students. Relative to other students in the study, students in District 1 scored around average (0.01 SD), while those in Districts 4 and 5 scored above average (0.21 and 0.07 SD, respectively), and those in Districts 2 and 3 scored below average (-0.25 and -0.22 SD, respectively) on the researcher-developed baseline mathematics assessment.

As we report elsewhere (Hill et al, 2015), instructional policies also differed across districts. Districts 1 and 2, which were located in the same state, both used the same set of NSF-funded curriculum materials, *Investigations in Number, Data, and Space* (TERC, 2008), which was developed to be aligned with the mathematics reforms promulgated in policy documents such as NCTM's *Principles and Standards of School Mathematics* (NCTM, 2000). Prior to the study, District 1 had further been engaged in an intensive, decade-long instructional improvement process centered on demanding mathematical pedagogy, featuring sustained teacher professional development and coaching. District 3 utilized a more traditional set of curriculum materials published by *Harcourt Brace*, and offered teachers relatively few opportunities for professional development. District 4 was engaged in implementing a new high-stakes teacher evaluation program during the study period. While District 4 also utilized an NSF-funded curriculum, *Everyday Mathematics* (UCSMP, 2007), and had a district mathematics coordinator, the district lacked a coordinated plan to improve elementary mathematics instruction. Finally, District 5 utilized the curriculum *Math Expressions*, generally considered to be more aligned with the reforms recommended in the NCTM *Standards* documents than the curriculum in District 3, but less so than the curricula used in Districts 1, 2, and 4. Over the course of study, the district

experienced a high degree of turnover in leadership, including the superintendent and math coordinator. At the same time, there were multiple efforts to provide reform-oriented professional development to teachers both during and prior to this study.

Data

Mathematical Quality of Instruction. The Mathematical Quality of Instruction (MQI) instrument (Hill et al., 2008), described above, evaluates teachers on instructional practices across two dimensions: *Ambitious Instruction* and *Errors and Imprecision*. Instrument developers originally envisioned four MQI dimensions, based largely on theory (Hill et al., 2008); however, factor analyses, as well as more recent substantive interpretations by instrument developers (Blazar et al, forthcoming), collapse these to two: *Ambitious Instruction* and *Errors and Imprecision*. For our analyses, we focus on teachers' *Ambitious Instruction* as opposed to teachers' *Errors and Imprecision* scores because the former dimension captures the level of inquiry-oriented instruction and activities that occur in the classroom (e.g., linking between multiple representations, solving a problem in multiple ways, student and teacher explanations, teachers' use of student ideas; see Figure 2).

Teacher *Ambitious Instruction* scores were generated from videotaped lessons of instruction captured over the course of two years. Teachers averaged 5.23 videotaped lessons (mode = 6), allowing for sufficient levels of predictive reliability (Hill et al., 2012).[1] Raters had a background in mathematics or mathematics education, passed a certification exam, and completed ongoing calibrations. Two raters scored teachers' instruction on each MQI item for each 7.5-minute lesson segment on a scale from 1 (*Low*) to 3 (*High*). We estimate two reliability statistics for *Ambitious Instruction.* First, we calculate the percentage of all instances in which the two raters assigned a teacher the same score on an item in a 7.5-minute lesson segment; this statistic gives

us a general measure of interrater reliability. Raters demonstrated exact agreement on 74% of all possible scoring instances. Second, we calculate the amount of variance in teacher scores attributable to the teacher (i.e., the intraclass correlation) as opposed to other sources of variation (i.e., lessons), adjusted for the modal number of lessons observed per teacher. Our estimate of 0.69 approximates conventionally acceptable levels (0.7), and is higher than those generated from similar studies (Bell et al., 2012; Kane and Staiger, 2012).

Given that teachers have differing numbers of lessons from which to construct *Ambitious Instruction* scores, we utilize empirical Bayes estimation to shrink scores back toward the mean based on their reliability (Raudenbush and Bryk, 2002). We calculate teacher-level scores by first averaging teacher performance on the MQI items within the *Ambitious Instruction* domain across segments and items. We then specify the following hierarchical linear model, where lessons are nested within teachers:

$$AMBITIOUS_{INSTRUCTION\,lj} = \mu_j + \varepsilon_{lj} \quad (1)$$

where μ_j is a random effect for teacher j, and ε_{lj} is a residual for each teacher's lesson. We utilize in analyses standardized estimates of the teacher-level random effect as the final MQI classroom observation score.

Student demographic and test score data. Most student data come from district administrative records, including student-teacher links from verified classroom rosters; student demographic information; and end-of-year mathematics and reading scores for standardized state tests completed in 2009, 2010, and 2011.

In addition, students completed a researcher-developed mathematics assessment, the *Upper-elementary Mathematics Assessment Modules* (Hickman et al., 2012), at the beginning and end of the school year.[2] This assessment was developed through a joint venture between Harvard

University and the Educational Testing Service, and is designed to be aligned with the Common Core State Standards for Mathematics. We describe these tests in more detail below.

Analyses

Our analysis is organized as follows. First, we conducted a quantitative analysis of the relationships between teachers' MQI scores and student achievement scores across districts. As we discuss below, the quantitative modeling revealed substantial differences among districts in the relationships between MQI and student achievement outcomes when using state standardized tests, but not when using the common researcher-developed assessment.

Next, we examined whether features of the state standardized tests that were used in different districts might contribute to the observed differences. Specifically, we compared all of the state tests as well as the researcher-developed assessment on several key dimensions of potential importance for the relationship with MQI scores, which we describe below. We present results concerning whether cross-test differences in test demand may be contributing to the differences in the relationships between teacher observation scores and student achievement outcomes.

Our methodology for exploring possible reasons for the observed cross-district differences is similar to that of Papay (2011), in that we are able to evaluate whether the pattern of results in our data is logically consistent with our hypotheses. With the limited number of districts, however, our design does not allow us to definitively test for the impact of each element. Given this limitation, we consider our findings suggestive of possible explanations for the variability in relationships between student test scores and teacher observation scores, rather than definitive.

Quantitative modeling. Other similar studies quantitatively test the relationship between teacher observation scores and student achievement scores by calculating a value-added score for each teacher, and then correlating this with the teacher's observation score. We take a slightly

different approach by including the observation score in our value-added model. This allows us to test formally for differences in the relationship between MQI and student achievement across districts. We estimate the following equation:

$$A_{jt} = \sum_{d=1}^{5} \beta_d MQI_k * D_d + \zeta f(A_{jt-1}) + \sigma X_{jt} + \theta C_{ct} + \alpha S_{sgt} + \pi G_{gt} + \varphi D_d + \mu_k + \epsilon_{jt} \qquad (2)$$

where the outcome of interest, A_{jt}, represents math score for student *j* at time *t*. We regress our outcome of interest on interactions between each teacher *k*'s MQI score on *Ambitious Instruction* and district *d*, $(MQI_k * D_d)$; a function of student's prior achievement, $f(A_{jt-1})$; a vector of student demographic variables, X_{jt}, including gender, race, free- or reduced-price lunch eligibility, special education status, and limited English proficiency; vectors of student demographic and test score variables aggregated to the class, C_{ct}, and school-grade, S_{sgt}, levels; grade-by-year fixed effects, G_{gt}, that account for different scaling of tests across grades and years; and district fixed effects, φD_d.[3] Given the nested structure of the data, we include a random effect for teachers, μ_k, as well as a student-level error term, ϵ_{jt}. We fit models using all years of available test score data to increase the precision of our estimates (e.g., Goldhaber and Hansen, 2012; Schochet and Chiang, 2013). We also limit the sample to those classes where fewer than 50% of students have special education status, where 50% or fewer are missing scores for the prior achievement vector, and, after all other restrictions, where there are at least five students. We perform these restrictions in order to exclude atypical classrooms from our value-added model; the restriction removes 5% of classrooms, and results in an average class size of approximately 18 students.

Our parameters of interest are in the vector β_d, which estimate the relationship between teachers' MQI score and student achievement for each district. To test whether these parameters differ across districts, we conduct a series of post-hoc general linear hypothesis (GLH) tests that compare all pairwise relationships. We fit this model separately for the researcher-developed assessment and the state standardized math tests.

Exploring variability in MQI/achievement relationships across districts.

***Test coding*.** In order to explore whether our data are consistent with the hypothesis that cross-test differences may be contributing to the cross-district differences in the relationships between teacher observation scores and student achievement outcomes, we coded state standardized math tests from each of the sampled districts and grades (fourth and fifth), as well as the researcher-developed test. We completed the test coding as follows.

First, we gathered information about the state test administered in each district from publicly accessible websites. Because Districts 1 and 2 come from the same state, test information and materials are the same. As we were able to recover test information from multiple years, and investigation of state test blueprints suggested that characteristics of each test largely remained constant from year to year, we randomly selected to analyze all the test items from a single school year. Of the four state tests, only the state tests from District 5 were publicly available in their complete and original form. For the other three state tests, we included in our analyses all available publicly released items in the school year analyzed,[4] and cross-referenced these against state test blueprints. We examined 95 items, on average, from each state test, across both grades. For the researcher-developed assessment, we randomly selected one year and one form from each of the sampled grades and coded these tests in their entirety.

The authors jointly coded these tests on two dimensions: *item format* (AERA/APA/NCME, 1999); and *alignment to the MQI instrument* (see Online Appendix A for a complete description of these coding schemes and procedures). To operationalize degree of alignment to the MQI, for each test, we assessed whether each test item demanded a high, medium, or low level of student engagement with MQI *Task Cognitive Activation,* which captures the degree of difficulty and challenge associated with a task. For example, items rated *low* included items which asked students to recall and apply procedures or to reproduce known facts or formulas. Items rated *high* asked students to engage with content at a high level of cognitive activation, such as by determining the meaning of mathematical concepts or relationships or drawing connections among different representations or concepts. To measure item format, we utilized the test item format categories described by the AERA/APA/NCME *Standards for Educational and Psychological Testing* (1999) to code the proportion of items on each test that were multiple choice, short answer (e.g., constructed response items; items which asked students to 'bubble-in' an answer), and open-ended (e.g., items requesting longer responses, short essays, or explanations of answers). To summarize these analyses, we created a summary metric for *overall test demand*. Each assessment was assigned a score of High, Mid, or Low on each dimension, based on our coding. The overall test demand codes reflected our holistic judgment of the overall levels of expected skills placed by the student achievement tests on the dimensions above.

Before beginning this process, the authors individually reviewed coding manuals, scored a subset of items from an external test, and held meetings in order to calibrate with one another over discrepant scores. In the final coding process, the authors coded all items jointly. We examined all items as a group, and only moved to the next item once agreement was reached on all codes. We note that during the coding process, the authors were not blind to the state test

from which items were derived. In order to minimize bias that may have resulted from this knowledge, we frequently compared our scores for specific items with the scores we assigned to similar items on different tests, checking for alignment.

Results

We begin by presenting results from our quantitative analysis, in which we examine cross-district differences in the relationships between MQI and student achievement on both the researcher-developed and state standardized tests. Next, we present results from our exploratory analyses examining potential factors contributing to the observed cross-district variability in relationships, including cross-district variability in the distribution of observed instructional quality and reported test prep behaviors, and differences in expected skills across student achievement tests.

Quantitative Models

In Table 3, we present estimates of the relationships between MQI and student achievement on both the researcher-developed and state standardized tests, for each district. In the first column, we present results for the researcher-developed assessment. We observe that the relationships between MQI *Ambitious Instruction* and student achievement on the researcher-developed assessment are generally small and not statistically significant.

In the second column of Table 3, we present results for the state standardized assessments. Comparing the second column with the first column, we observe that, compared with the results for the researcher-developed assessment, results for the relationship between MQI observation scores and student performance on state standardized assessments are more varied across districts – with estimates ranging from -0.053 in District 4 to 0.126 in District 2. The relationship

between MQI *Ambitious Instruction* scores and student achievement on state standardized tests was significantly different from zero only in Districts 1 and 2 (which are in the same state).

In Table 4, we present the results of a series of post-hoc GLH tests comparing the coefficients from the models in Table 3 across districts. As seen in the first column, on the researcher-developed mathematics test, we observe only one statistically significant difference in the MQI/student achievement relationship across districts – between Districts 2 and 5 (p=.03). By contrast, examining the second column, we observe more cross-district differences using the state standardized assessments. Note that we do not find differences in the relationship between MQI scores and student achievement outcomes on the state test in District 1 versus in District 2. Recall that these two districts are in the same state, hence students take the same state test. By contrast, the relationships between MQI and student achievement on the state standardized tests are stronger in District 1 than in Districts 3 and 4 (p=.08 and 0.01, respectively). In addition, the relationships between MQI and student achievement are stronger in District 2 than in Districts 3, 4, and 5 (p=.02, <.01, and .08, respectively).

Differences in Expected Skills across Student Achievement Tests.

To summarize the results from our test coding and analyses of the differences in expected skills across student achievement tests, we present results from a summary metric for overall test demand (see Table 4). The results from our coding of the dimensions that contributed to the summary measure are found in Online Appendix B.

Overall, we judged the state test utilized in Districts 1 and 2 to be highest in demand, followed by the researcher-developed assessment. We judged the state test in Districts 3 to be of moderate demand, and those in Districts 4 and 5 to be of moderate to low demand. On the one hand, these results seem somewhat consistent with the hypothesis that stronger relationships

between teacher observation scores and student test scores may be observed when characteristics of the tests used to measure student achievement are more aligned with characteristics of the observation instruments. As noted earlier, we observed the strongest relationships between MQI scores and student achievement in Districts 1 and 2, where the demand of the state test was relatively high. In Districts 3 through 5, where the overall demand of the state tests was relatively lower, we generally observed weaker relationships.

However, when student achievement is measured using the researcher-developed assessment, the relationship between the MQI and student achievement is significantly different between Districts 2 and 5, despite the fact that the researcher-developed assessment presented the same demands to all students across districts. Consequently, variation in test characteristics does not appear to be solely responsible for the observed differences.

Discussion

In summary, we found that relationships between teachers' MQI observation scores and their students' achievement on state mathematics tests differed by district. We explored whether differential characteristics of the tests, or *test-observation misalignment*, might contribute to this variability. If alignment between test content and instructional quality drives these relationships, then we would expect to see two things: (a) variability in the relationship between MQI scores and student achievement across districts on state standardized tests, with stronger relationships in districts where state tests are better aligned to the MQI; and (b) no variability in the relationship between MQI scores and student achievement across districts on the researcher-developed assessment.

While these analyses are exploratory in nature, our findings do suggest some support for the test-observation misalignment hypothesis. We found that the relationships between teachers'

MQI scores and their students' achievement on state standardized tests were the strongest in Districts 1 and 2. These two districts also had the most demanding state standardized test, as rated on the summary metric of overall test demand. In particular, this test had the highest proportion of open-ended items in the sample, with nearly a quarter of all items coded as open-ended. This compared with few or no open-ended items on the other states' tests. In addition, although the average levels of cognitive demand for all state tests were relatively low, with most items coded as 'perform procedures,' the state test for Districts 1 and 2 was distinguished by the highest mean level of cognitive demand of the state tests in the sample.

By contrast, in Districts 3, 4, and 5, relationships between teachers' MQI scores and their students' state standardized test scores were smaller and not statistically significant. These districts also used state tests whose overall demand was judged to be middling to low. The state tests used in Districts 3, 4, and 5 demonstrated middling cognitive demand and alignment with the MQI instrument, and relied heavily on multiple choice items. The Districts 4 and 5 tests were entirely multiple choice.

At the same time, we also observed one cross-district difference in the relationship between MQI and student achievement on the researcher-developed assessment. Because the researcher-developed assessment presents the same demands of all students across districts, and the MQI presents the same demands of all teachers across districts, this data point is inconsistent with the notion that test characteristics could be the sole reason for variability in relationships.

Cross-district differences in relationships between teacher observation scores and student achievement: Potential alternative explanations

Considering the current findings in the broader district policy context, we speculate that contributing factors we discuss are likely intertwined: Cross-district differences in relationships

between teacher observation and student achievement scores may arise in part from the interplay between characteristics of state tests and the cross-district differences in teacher instructional quality, potentially spurred by how instruction is measured and implemented in school districts.

Elsewhere, we describe how measures of teachers' instructional quality, including their *Ambitious Mathematics Instruction*, differ substantively across the districts assessed in this study (Hill et al., 2015). Blazar and colleagues (2016) also found that teachers' value-added scores, or contributions to their students' math performance as measured on state standardized assessments, signaled different sets of mathematics-specific instructional practices across these districts. These patterns were not explained away by observable background characteristics of teachers, including their math content knowledge, prior coursetaking in mathematics, and certification pathway, suggesting that factors beyond labor market sorting and instead internal to district context likely played a key role.

These findings are consistent with prior research that has emphasized the important roles that local instructional policies, including testing policy, curriculum, and professional development opportunities, play in influencing teachers' instruction (e.g., Booher-Jennings, 2005; Coburn, 2005; Cohen and Hill, 2000). With regard to the relationship between high-stakes testing and instructional practices, for example, Booher-Jennings (2005) found that when confronted with a new high-stakes testing policy, teachers in one urban elementary school altered their instructional practices markedly to attempt to improve test scores, for example by providing targeted tutoring to students believed to be on the border of passing the test while requiring the rest of the class to perform seatwork.

As noted above, districts in the current study varied in their instructional policies, which may have both reflected and shaped teachers' differential responses to high-stakes testing across

districts. Recall that Districts 1 and 2, which were located in the same state, both used the same set of inquiry-oriented curriculum materials, and provided teachers with aligned professional development (especially so in District 1). In this context, we speculate that perhaps in Districts 1 and 2, for example, teachers may have been incented to provide ambitious instruction aligned both to their state's high-level demand test and to the MQI instrument, and provided with curricular and professional development resources to scaffold and support this challenging pedagogy. This interplay between facets of the test and the way in which instruction was implemented may have supported alignment between student achievement and teacher observation outcomes. On the other hand, in District 3, where a new high-stakes teacher evaluation program was being implemented, teachers may have had incentives to engage in more test prep-oriented instruction that was well aligned to their state's relatively basic skills-oriented test, but poorly aligned to the ambitious instructional goals valued in the MQI observation instrument. Here, properties of the state test may have interacted with the district policy climate to hinder alignment between student test and teacher observation scores.

Another possibility is that factors beyond the classroom, such as parent involvement, may both respond to instructional quality and influence test scores while also varying by district. For example, it is possible that parents could respond to weak instructional quality in their children's classrooms by supplementing with extra tutoring or afterschool programs (Hill et al., 2008). Such practices could attenuate the observed relationship between observed instructional quality and student achievement outcomes, and could also vary by district if the availability or quality of supplemental programs varies across districts. As noted above, however, these observations are speculative. Additional research is needed to parse how specific factors may be contributing to variability in alignment, as we discuss below.

Limitations

This study presents several limitations, which point to potential directions for future research. First, we are unable to empirically model and test the strength of test-observation misalignment to variability in the relationship between MQI and student achievement, as the dimensions we examine are perfectly collinear with district. Second, the current study is limited to the case of the MQI. As districts overhaul their teacher evaluation systems, ushering in new assessments and observation instruments, future research is needed to explore whether the observed relationships between characteristics of tests and observation scores are replicated across different observation instruments and contexts. In addition, further work is needed in order to parse in detail the district-level factors which mediate the relationships between test score and observation outcomes, which are likely overlapping and intertwined. For example, future studies should analyze districts' adopted curricula in depth for their alignment with inquiry-oriented instructional practices as valued in observation instruments, and for their cognitive demand, in order to determine how curriculum may moderate the relationship between classroom observations and test scores. In addition, complementary ethnographic studies that analyze cross-district differences in how math instruction and student achievement play out when embedded in tensions of policy could shed light on these issues.

Conclusions and Future Directions

In the current study, we find evidence that the relationship between instructional quality, as judged by classroom observations, and student achievement on state tests is different in different districts. We suggest that one possible factor which might contribute to variability in these relationships may be different sensitivity of the observation instrument to different state tests.

When observation instruments and student assessments are more aligned, stronger relationships between instructional quality and student achievement may result.

In a broader theoretical context, the current study raises questions about the goals of classroom observation instruments versus standardized achievement tests, what each type of assessment prioritizes as important, and what each offers and does not offer us as we work to understand and assess teacher quality. Specifically, the type of learning promoted on classroom observations versus in standardized tests may be in discord. The MQI and many other classroom observation instruments begin with the learning goals of fostering students' conceptual understanding and skill learning (Grossman et al., 2014). To assess teachers' effectiveness in providing students opportunities to meet these learning goals, classroom observation instruments often measure instructional practices which promote students' conceptual learning, in line with the conceptually-oriented instructional practices identified in the mathematics education literature (Hiebert and Gruows, 2007). By contrast, while state standardized tests are also often *intended* to evaluate students' progress toward both conceptual and skill efficiency learning goals, as envisioned in state standards (Webb, 1999), in reality they often disproportionately measure lower-level skills (Resnick et al., 2004). Evaluating teachers using student standardized test scores may thus result in high evaluation scores for teachers using skill efficiency-oriented instruction, while not rewarding teachers utilizing more conceptually-oriented instruction for the relative affordances they are offering to students' conceptual learning and richer procedural understanding.

How might the current study's findings be applicable for the work of states and districts as they seek to implement more rigorous curriculum standards, such as the Common Core? On the one hand, the implementation of more rigorous curriculum standards such as the Common Core

seems poised to streamline the content of student assessments, as states and districts move to adopt new assessments (e.g., Partnership for Assessment of Readiness in College and Careers; Smarter Balanced Assessment Consortium) designed to be aligned with more rigorous content standards (Grossman et al., 2011). These assessments are indeed being designed with the goal of improving upon some of the shortcomings of the current state tests that we noted in the current study, such as disproportionate emphasis on lower-level skills and the multiple choice format. To the extent that these new assessments succeed in their goals, a problem that we detect in the current study – namely, the misalignment between the lower-level skills often demanded on student tests and the higher-level skills teachers are expected to foster on new observation instruments – may be minimized. However, it is not clear to date the extent to or speed with which states will move to replace their existing local tests with the new, more conceptually-oriented assessments. Indeed, a report from the National Governors Association Center for Best Practices expresses concern that states may simply tack on new assessments to their existing testing regimens, resulting in over-testing while retaining some of the problems associated with the current tests (Grossman et al., 2011).

In either case, concerns over the alignment between test-based measures of effectiveness and teacher observations remain pressing for policy. While perfect alignment between these measures is not expected, varying relationships may prove problematic for teacher evaluation policy and practice. For example, under current conditions, a teacher who focuses her instruction on lower-level tasks in anticipation of a basic skills-oriented state standardized test could potentially receive a conflicting evaluation report, indicating high value-added on a state standardized test, coupled with low teacher observation scores. In such a case, the teacher may face unclear guidance on how to improve her practice. The current results suggest that districts

may need to examine their classroom observation instruments for alignment with their high-stakes student assessments, in order to aid both teachers and districts in better supporting teachers for instructional effectiveness.

Notes

[1] Teachers were allowed to select the dates for videotaping in advance. Project managers required that teachers select a typical lesson and exclude days on which students were taking a test. While it is possible that these videotaped lessons are unique from a teachers' general instruction, prior research suggests that when teachers were given discretion to choose their 'best' classroom videos from a set, the chosen videos provided essentially similar information about teachers' instructional quality as the videos that were not chosen (Ho and Kane, 2013).

[2] In District 4, teachers began the overarching study in the second semester of the 2010-2011 school year. Because characteristics of the tests themselves are a major component of our analyses, we exclude student-level administrative data from District 4 in this school year from our analyses. We use complete student-level administrative data from the district in the 2011-2012 school year.

[3] We also consider a fully interacted model that allows all parameters on covariates to vary by district. However, we find that results follow the same pattern and, therefore, use this more parsimonious model.

[4] The exception to this was the test from District 4, for which we included all publically available items that we could recover, regardless of year, due to the comparatively lower number of released items.

References

American Educational Research Association/American Psychological Association/ National Council on Measurement in Education. (1999). *Standards for educational and psychological testing*. Washington, DC: AERA.

Bell, C., Gitomer, D., McCaffrey, D., Hamre, B., Pianta, R, and Qi, Y. (2012). An argument approach to observation protocol validity. *Educational Assessment, 17*(2-3), 62-87.

Blazar, D. (2015). Effective teaching in elementary mathematics: Identifying classroom practices that support student achievement. *Economics of Education Review, 48*, 16-29.

Blazar, D., Braslow, D., Charalambous, C. Y., and Hill, H. C. (forthcoming). Attending to general and mathematics-specific dimensions of teaching: Exploring factors across two observation instruments. *Educational Assessment*.

Blazar, D., Litke, E., and Barmore, J. (2016). What does it mean to be ranked a "high" or "low" value-added teacher? Observing differences in instructional quality across districts. *American Educational Research Journal, 53*(2), 324-359.

Booher-Jennings, J. (2005). Below the bubble: "Educational triage" and the Texas accountability system. *American Educational Research Journal*, *42*(2), 231-268. doi: 10.3102/00028312042002231

Borko, H., Eisenhart, M., Brown, C. A., Underhill, R. G., Jones, D., and Agard, P. C. (1992). Learning to teach hard mathematics: Do novice teachers and their instructors give up too easily?. *Journal for Research in Mathematics Education*, *23*(3), 194-222. doi: 10.2307/749118

Boston, M. (2012). Assessing instructional quality in mathematics. *The Elementary School Journal*, *113*(1), 76-104.

Coggshall, J. G., Rasmussen, C., Colton, A., Milton, J., and Jacques, C. (2012). *Generating teacher effectiveness: The role of job-embedded professional learning in teacher evaluation.* Washington, DC: National Comprehensive Center for Teacher Quality.

Coburn, C. E. (2005). The role of nonsystem actors in the relationship between policy and practice: The case of reading instruction in California. *Educational Evaluation and Policy Analysis, 27,* 23-52. doi: 10.3102/01623737027001023

Cohen, D. K. (1990). A revolution in one classroom: The case of Mrs. Oublier. *Educational Evaluation and Policy Analysis, 12,* 311–330.

Cohen, D. K. (2011). *Teaching and its predicaments.* Cambridge, MA: Harvard University Press.

Cohen, D. K., and Hill, H. C. (2000). Instructional policy and classroom performance: The mathematics reform in California. *Teachers College Record*, *102*(2), 294-343. doi: 10.1111/0161-4681.00057

Cohen, D. K., Raudenbush, S. W., and Ball, D. L. (2003). Resources, instruction, and research. *Educational Evaluation and Policy Analysis*, *25*(2), 119-142. doi: 10.3102/01623737025002119

Daley, G., and Kim, L. (2010). *A teacher evaluation system that works*. Santa Monica, CA: National Institute for Excellence in Teaching.

Danielson, C. (2011). *Enhancing professional practice: A framework for teaching*. ASCD.

Doyle, W. (1988). Work in mathematics classes: The context of students' thinking during instruction. *Educational Psychologist*, *23*(2), 167-180.

Firestone, W. A. (2014). Teacher evaluation policy and conflicting theories of motivation. *Educational Researcher*. doi: 0013189X14521864

Gallagher, H. A. (2004). Vaughn Elementary's innovative teacher evaluation system: Are teacher evaluation scores related to growth in student achievement? *Peabody Journal of Education*, *79*(4), 79-107. doi: 10.1207/s15327930pje7904_5

Gamse, B. C., Jacob, R. T., Horst, M., Boulay, B., and Unlu, F. (2008). *Reading First impact study*. Washington, DC: Institute of Education Sciences, U.S. Department of Education.

Goldhaber, D., and Hansen, M. (2012). Is it just a bad class? Assessing the long-term stability of estimated teacher performance. *Economica, 80*(319), 589-612. doi: 10.1111/ecca.12002

Good, T., and Brophy, J. (2007). *Looking in classrooms.* New York: Harper Collins.

Grossman, P., Cohen, J., Ronfeldt, M., and Brown, L. (2014). The test matters: The relationship between classroom observation scores and teacher value added on multiple types of assessment. *Educational Researcher*, *43*(6), 293-303. doi: 10.3102/0013189X14544542

Grossman, T., Reyna, R., and Shipton, S. (2011). Realizing the potential: How governors can lead effective implementation of the Common Core State Standards. National Governors Association Center for Best Practices.

Harris, D. N., Ingle, W. K., and Rutledge, S. A. (2014). How teacher evaluation methods matter for accountability: A comparative analysis of teacher effectiveness ratings by principals and teacher value-added measures. *American Educational Research Journal*, *51*(1), 73-112. doi: 0002831213517130.

Henningsen, M., and Stein, M. K. (1997). Mathematical tasks and student cognition: Classroom-based factors that support and inhibit high-level mathematical thinking and reasoning. *Journal for Research in Mathematics Education*, 524-549.

Hickman, J. J., Fu, J., and Hill, H. C. (2012). *Technical report: Creation and dissemination of upper-elementary mathematics assessment modules.* Princeton, NJ: Educational Testing Service.

Hill, H. C., Blazar, D., and Lynch, K. (2015). Resources for teaching: Examining personal and institutional predictors of high-quality instruction. *AERA Open, 1*(4), 1-23.

Hill, H. C., Blunk, M. L., Charalambous, C. Y., Lewis, J. M., Phelps, G. C., Sleep, L., and Ball, D. L. (2008). Mathematical knowledge for teaching and the mathematical quality of instruction: An exploratory study. *Cognition and Instruction*, *26*(4), 430-511. doi: 10.1080/07370000802177235

Hill, H. C., Charalambous, C. Y., Blazar, D., McGinn, D., Beisiegel, M., Humez, A., Kraft, M., Litke, E., and Lynch, K. (2012). Validating arguments for observational instruments: Attending to multiple sources of variation. *Educational Assessment, 17*(2-3), 88-106.

Hill, H. C., Charalambous, C. Y., and Kraft, M. A. (2012). When rater reliability is not enough: Teacher observation systems and a case for the generalizability study. *Educational Researcher, 41*(2), 56-64. doi: 10.3102/0013189X12437203

Hill, H. C., Kapitula, L. R., and Umland, K. L. (2011). A validity argument approach to evaluating value-added scores. *American Educational Research Journal*, *48*, 794–831. doi: 10.3102/0002831210387916

Hill, H. C., Rowan, B., and Ball, D. L. (2005). Effects of teachers' mathematical knowledge for teaching on student achievement. *American Educational Research Association, 42*(2), 371-406. doi: 10.3102/00028312042002371

Ho, A. D., and Kane, T. J. (2013). *The reliability of classroom observations by school personnel*. Seattle: The Bill and Melinda Gates Foundation.

Jacob, B. (2005). Accountability, incentives and behavior: Evidence from school reform in Chicago. *Journal of Public Economics, 89*(5-6), 761-796. doi: 10.1016/j.jpubeco.2004.08.004

Kane, T., and Cantrell, S. (2010). Learning about teaching: Initial findings from the measures of effective teaching project. *MET Project Research Paper, Bill & Melinda Gates Foundation, 9*.

Kane, T. J., McCaffrey, D. F., Miller, T., and Staiger, D. O. (2013). *Have we identified effective teachers? Validating measures of effective teaching using random assignment.* Seattle: The Bill and Melinda Gates Foundation.

Kane, T. J., and Staiger, D. O. (2012). *Gathering feedback for teaching: Combining high-quality observations, student surveys, and achievement gains.* Seattle, WA: The Measures of Effective Teaching Project, The Bill and Melinda Gates Foundation.

Kane, T. J., Taylor, E. S., Tyler, J. H., and Wooten, A. L. (2011). Identifying effective classroom practices using student achievement data. *Journal of Human Resources, 46*(3), 587-613.

Koedel, C., and Betts, J. R. (2011). Does student sorting invalidate value-added models of teacher effectiveness? An extended analysis of the Rothstein critique. *Education, 6*(1), 18-42. doi: 10.1162/EDFP_a_00027

Konstantopoulos, S. (2014). Teacher effects, value-added models, and accountability. *Teachers College Record, 116*(1), 1-21.

Ma, L. (1999). *Knowing and teaching elementary mathematics*. Mahwah, NJ: Lawrence Erlbaum Associates.

McCaffrey, D. F., Sass, T. R., Lockwood, J. R., and Mihaly, K. (2009). The intertemporal variability of teacher effect estimates. *Education Finance and Policy, 4*(4), 572-606. doi:

10.1162/edfp.2009.4.4.572

Milanowski, A. (2004). The relationship between teacher performance evaluation scores and student achievement: Evidence from Cincinnati. *Peabody Journal of Education, 79*(4), 33-53. doi: 10.1207/s15327930pje7904_3

Milanowski, A. (2011). *Validity research on teacher evaluation systems based on the Framework for Teaching.* Madison, WI: Consortium for Policy Research in Education.

National Governors Association Center for Best Practices, Council of Chief State School Officers (2010). *Common Core State Standards for Mathematics*. Author: Washington, D.C.

National Research Council. (2001). *Adding it up: Helping children learn mathematics*. Washington, DC: National Academy Press.

NCTM. (1995). *Assessment standards for school mathematics*. Reston, VA: Author.

NCTM. (2000). *Principles and standards for school mathematics*. Reston, VA: NCTM.

Papay, J. P. (2011). Different tests, different answers: The stability of teacher value-added estimates across outcome measures. *American Educational Research Journal, 48*(1), 163-193. doi: 10.3102/0002831210362589

Pianta, R. C., Hamre, B. K., and Mintz, S. (2010). *Classroom Assessment Scoring System (CLASS) Manual: Upper Elementary*. Teachstone.

Podgursky, M. J., and Springer, M. G. (2007). Teacher performance pay: A review. *Journal of Policy Analysis and Management, 26*(4), 909. doi: 10.1002/pam.20292

Polikoff, M. S. (2014). Does the test matter? Evaluating teachers when tests differ in their sensitivity to instruction. In T. J. Kane, K. A. Kerr, and R. C. Pianta (Eds.), *Designing*

Teacher Evaluation Systems: New Guidance from the Measures of Effective Teaching Project (pp. 278-302). San Francisco, CA: Jossey-Bass.

Porter, A. C. (2002). Measuring the content of instruction: Uses in research and practice. *Educational Researcher, 31*(7), 3-14. doi: 10.3102/0013189X031007003

Raudenbush, S. W., and Bryk, A. S. (2002). *Hierarchical linear models: Applications and data analysis methods. Second Edition*. Thousand Oaks, CA: Sage Publications.

Resnick, L. B., Rothman, R., Slattery, J. B., and Vranek, J. L. (2004). Benchmarking and alignment of standards and testing. *Educational Assessment, 9*(1-2), 1-27. doi: 10.1080/10627197.2004.9652957

Richland, L. E. (2015). Linking gestures: Cross-cultural variation during instructional analogies. *Cognition and Instruction, 33*(4), 295-321.

Sawada, D., Piburn, M. D., Judson, E., Turley, J., Falconer, K., Benford, R., and Bloom, I. (2002). Measuring reform practices in science and mathematics classrooms: The reformed teaching observation protocol. *School Science and Mathematics, 102*(6), 245-253.

Schacter, J., and Thum, Y. M. (2004). Paying for high- and low-quality teaching. *Economics of Education Review, 23*(4), 411-430. doi: 10.1016/j.econedurev.2003.08.002

Schochet, P. Z., and Chiang, H. S. (2013). What are error rates for classifying teacher and school performance using value-added models? *Journal of Educational and Behavioral Statistics, 38*(2), 142-171. doi: 10.3102/1076998611432174

Schoenfeld, A. H. (2004). The math wars. *Educational Policy, 18*(1), 253-286.

Spearman, C. (1904). The proof and measurement of association between two things. *American Journal of Psychology, 15*, 72–101.

Stein, M. K., Baxter, J. A., and Leinhardt, G. (1990). Subject-matter knowledge and elementary instruction: A case from functions and graphing. *American Educational Research Journal*, *27*(4), 639–663.

Stein, M. K., Engle, R. A., Smith, M. S., and Hughes, E. K. (2008). Orchestrating productive mathematical discussions: Five practices for helping teachers move beyond show and tell. *Mathematical Thinking and Learning*, *10*(4), 313-340. doi: 10.1080/10986060802229675

Stein, M. K., Grover, B. W., and Henningsen, M. (1996). Building student capacity for mathematical thinking and reasoning: An analysis of mathematical tasks used in reform classrooms. *American educational research journal*, *33*(2), 455-488.

Stein, M. K., and Lane, S. (1996). Instructional tasks and the development of student capacity to think and reason: An analysis of the relationship between teaching and learning in a reform mathematics project. *Educational Research and Evaluation*, *2*, 50–80.

Stevens, F. I. (1993). Applying an opportunity-to-learn conceptual framework to the investigation of the effects of teaching practices via secondary analyses of multiple-case-study summary data. *Journal of Negro Education*, *62*(3), 232-248. doi: 10.2307/2295463

Stigler, J. W., and Hiebert, J. (1999). The teaching gap: Best ideas from the world's teachers for improving education in the classroom. New York: Free Press.

TERC. (2008). *Investigations in number, data, and space*. Pearson Scott Foresman.

University of Chicago School Mathematics Project (UCSMP). *Everyday Mathematics.* McGraw-Hill, 2007.

Weisberg, D., Sexton, S., Mulhern, J., and Keeling, D. (2009). The widget effect: Our national failure to acknowledge and act on differences in teacher effectiveness. Brooklyn, NY: New Teacher Project.

Yackel, E., and Cobb, P. (1996). Sociomathematical norms, argumentation, and autonomy in mathematics. *Journal for Research in Mathematics Education, 27*(4), 458-477.

Tables

Table 1

Elements included in the Mathematical Quality of Instruction (MQI), Ambitious Instruction Factor.

Dimension	Description of Teaching Practices	Example Items
Working with Students and Mathematics	Captures the extent to which teachers interpret and respond to students' mathematical productions, including utterances, written work, and mathematical errors.	*Responding to student mathematical productions,* such as by building instruction around students' strategies or ideas, or identifying and distilling mathematical points from students' utterances or work. *Remediation of student errors and difficulties*: Substantially addressing students' misconceptions and difficulties with the mathematical content.
Student Participation in Meaning-Making and Reasoning	Captures the extent to which there is evidence in the classroom of students' involvement in cognitively activating work.	*Engaging in reasoning,* such as by identifying and explaining patterns, or making connections across multiple representations, concepts, or solution strategies. *Making mathematical claims or counterclaims, or asking mathematically motivated questions.* *Providing explanations*.
Richness of the Mathematics	Captures the depth of the mathematics students are provided the opportunity to learn, with attention to the meaning of facts, procedures, and concepts or to important mathematical practices.	*Explanations*: Providing mathematical meaning to facts, procedures, concepts, or solution strategies. *Multiple procedures or solution methods*: Comparing or considering multiple solution strategies for a problem. *Linking and connections*: Drawing explicit

	links among concepts, representations, and procedures. *Developing generalizations*: Building generalizations of mathematical facts, procedures, or concepts using specific examples. *Mathematical language*: Using dense and precise mathematical language.

Note: For a detailed discussion of the instrument and items, see Hill et al., 2008.

Table 2

Sample Descriptive Statistics

	District 1	District 2	District 3	District 4	District 5
Teachers					
Male	29.23	13.46	13.64	9.20	0
African-American	20.00	1.92	59.09	18.68	11.11
White	63.08	80.77	29.55	70.33	74.07
Other (Asian, Hispanic)	15.15	9.43	7.14	4.35	14.81
Experience	10.64	10.44	8.72	11.69	8.26
Traditionally Certified	74.24	86.79	47.62	90.22	96.30
Alternatively Certified	6.06	0	23.81	4.35	3.70
No Certification	10.61	7.55	16.67	1.09	0
Observations	66	53	44	92	27
Students					
Male	47.79	51.80	46.48	49.97	49.73
African-American	37.46	51.95	66.51	31.22	57.85
Asian	13.27	3.42	2.38	9.41	3.35
Hispanic	37.35	12.90	11.80	21.69	6.52
White	6.70	27.44	17.64	33.79	23.81
Eligible for Free- or Reduced-Price Lunch	82.19	72.70	61.26	49.07	72.08
Special Education Status	12.90	11.65	9.65	9.41	9.17
Limited English Proficient	36.43	23.21	10.01	11.74	0.71
Researcher-Developed Assessment Baseline Achievement	0.01	-0.25	-0.22	0.21	0.07
Observations	1,628	2,077	839	1,669	567

Note: Teacher characteristics generated from a sub-sample of those who completed a baseline survey.

Table 3

Estimates of the Relationship Between Instructional Quality on the measure of Ambitious Instruction and Student Achievement Across Districts

	Researcher-Developed Mathematics Assessment	State Standardized Tests
District 1	0.035	0.081**
	(0.024)	(0.031)
District 2	0.095*	0.126**
	(0.039)	(0.041)
District 3	0.046	-0.004
	(0.040)	(0.039)
District 4	0.021	-0.053
	(0.032)	(0.040)
District 5	-0.015	0.028
	(0.030)	(0.038)

Notes: ~ p<.10, * p<.05, ** p<.01, ***p<.001. Each column represents a separate regression model. Robust standard errors in parentheses. Sample includes 298 teachers and 6,780 students.

Table 4

Estimates of the Relationship Between Instructional Quality on the measure of Ambitious Instruction and Student Achievement Across Districts

P-values on post-hoc GLH tests		
All District Coefficients Equal	0.40	**0.02**
District 1 = District 2	0.20	0.39
District 1 = District 3	0.82	**0.08**
District 1 = District 4	0.73	**0.01**
District 1 = District 5	0.20	0.30
District 2 = District 3	0.39	**0.02**
District 2 = District 4	0.14	**<.01**
District 2 = District 5	**0.03**	**0.08**
District 3 = District 4	0.64	0.38
District 3 = District 5	0.21	0.57
District 4 = District 5	0.40	0.15

Notes: Sample includes 298 teachers and 6,780 students. *P*-values below 0.10 are bolded.

Table 5
Overall Test Demand

	Test Characteristics		Overall Demand
	Item Format	Alignment to MQI – Task Cognitive Activation	
State Tests			
Districts 1 and 2	High	High	High
District 3	Mid	Mid	Mid
District 4	Low	Mid	Mid/Low
District 5	Low	Mid	Mid/Low
Common Assessment			
All Districts	Mid	High	High/Mid

Figures

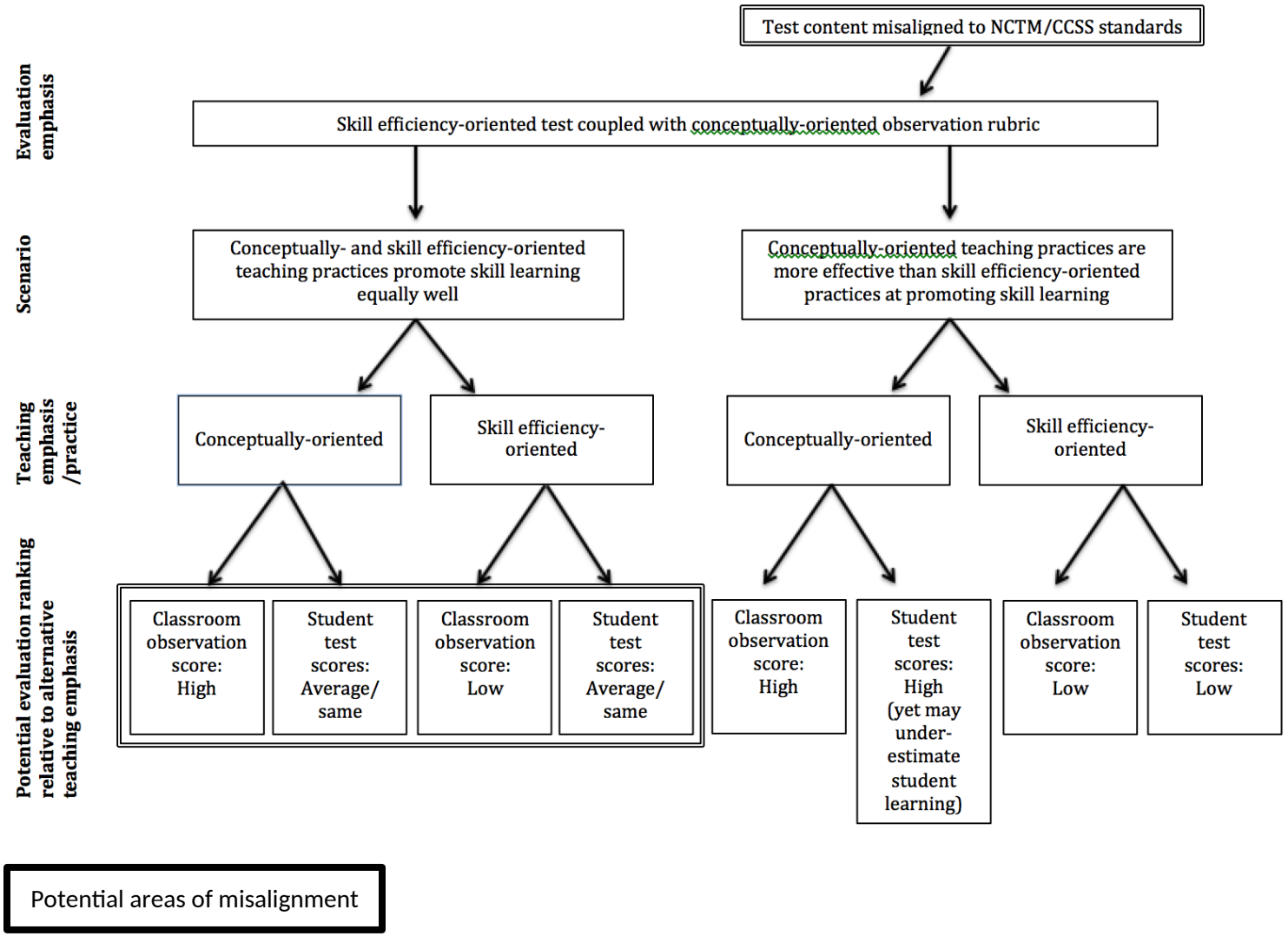


Figure 1. Logic model for hypothesized relationship between instructional practices captured by classroom observation and student learning as measured by assessments.

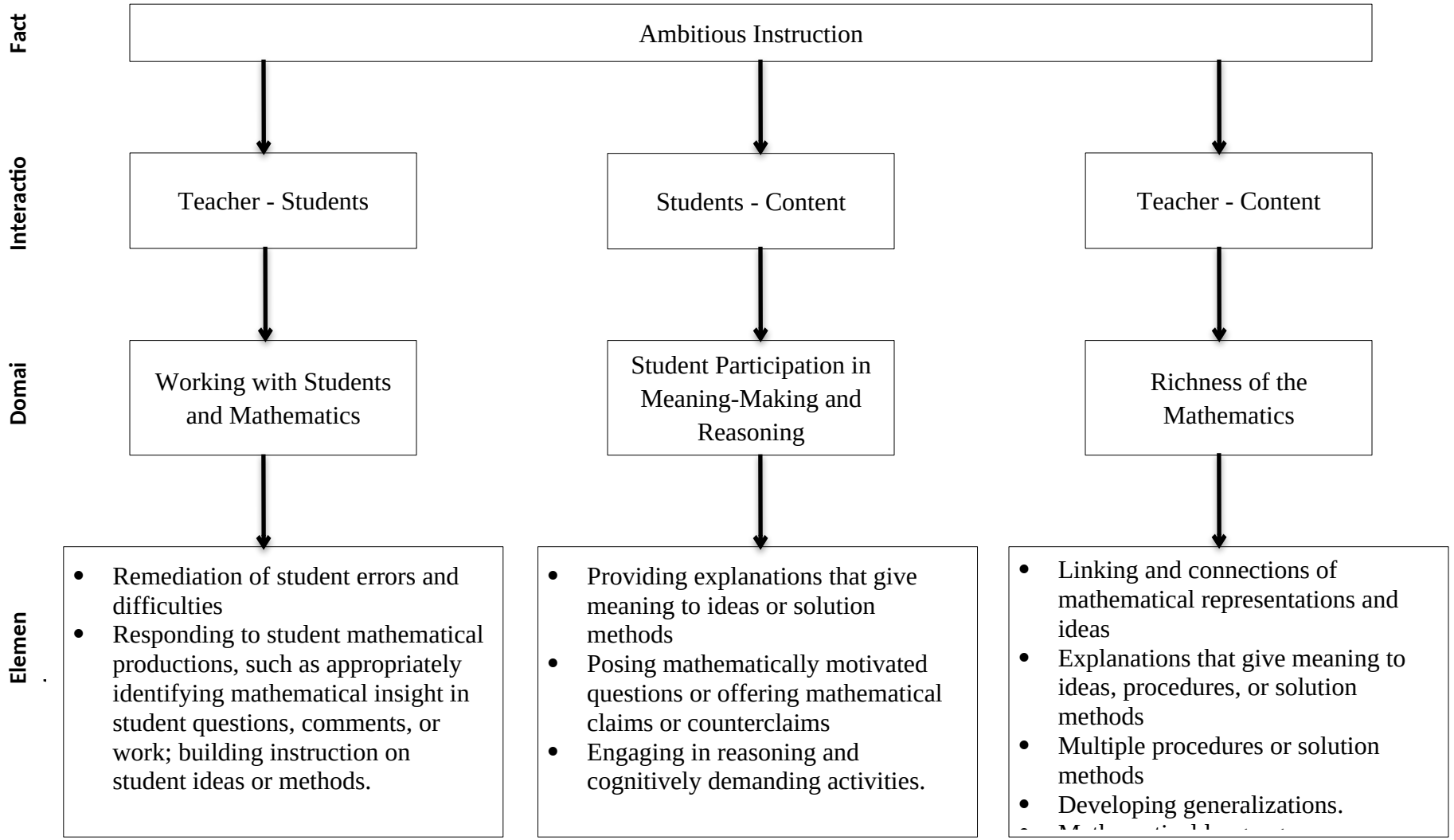


Figure 2. MQI Instructional Dimensions for the *Ambitious Instruction* Factor.

Online Appendix A:
Description of Test Coding

Item Format

One possible contributor to the cognitive demand of student assessments is item format. If the level of cognitive demand presented by a test is substantially lower than the cognitive demand of tasks teachers typically use in instruction, teachers may receive high scores on an observation instrument, while their students receive average scores on a test which allows them to display their skill efficiency, but not their conceptual learning gains. Low cognitive demand of a student test may thus weaken the relationship between classroom observation scores and student achievement.

To develop a picture of each test's distribution of item formats, we utilized the test item format categories described by the AERA/APA/NCME *Standards for Educational and Psychological Testing* (1999) to code the proportion of items on each test that were multiple choice, short answer (including constructed response items and items in which students were asked to 'bubble-in' an answer), and open-ended (such as items requesting longer responses, short essays, or explanations of answers).

Alignment with MQI Instrument: Task Cognitive Activation

Previous research has found that the cognitive demand of state tests is often relatively low, and that state test items tend to emphasize procedural skills and lower-level cognitive demand tasks (Resnick et al., 2004; Webb, 1999). For example, in a study of four state assessments in mathematics and three in science, Webb (1999) found that a large proportion of test items tested content at a lower level of cognitive demand than that indicated in the associated curriculum standards. In another study analyzing five state assessments in mathematics and English language arts, Resnick and colleagues (2004) found that in general, test items on state assessments were inappropriately easy relative to the levels of cognitive demand indicated in state curriculum standards. An analysis of four state English Language Arts assessments by Wixson and colleagues (2002) similarly found that the cognitive complexity of state assessment tasks tended to be lower than that of the relevant objectives. If the level of cognitive demand asked of the students in engaging with the content on a test is substantially lower than the demand of tasks incorporated in their teachers' instruction, the relationship between instructional quality and student achievement scores may be weakened.

We examined the extent of the alignment between the MQI instrument and the skills and competencies measured by the various student assessments. We asked: Did the skills and competencies valued by the MQI observation instrument align with the skills and competencies that the test items demanded?

To operationalize the degree of alignment, for each test, we assessed whether each test item demanded a high, medium, or low level of student engagement with the MQI dimension *Task Cognitive Activation* – scored on a scale for low (1) to high (3). This dimension represents a summary measure of students' opportunities to engage with rich and cognitively activating mathematics, as defined by the MQI instrument, and we hypothesized that it could be captured in a student assessment. Conversely, other MQI dimensions are important but do not map readily onto skills and competencies demanded on a student assessment, and thus were not coded for in the current analysis. For example, the MQI dimension of *Errors and Imprecision* captures the number and severity of mathematical errors that teachers make during instruction. While this dimension seems quite important to capture on an observational measure of teachers' instruction, we judged that this did not map easily to an analogous category of skills and competencies that

we could examine on student assessments. Note that in the analysis presented in the main body of the paper above, we used scores derived from a factor analysis, which indicated that multiple MQI dimensions loaded onto a single factor, which we termed *Ambitious Instruction*. By contrast, we conducted test coding using the original MQI dimensions.

As a sensitivity analysis, we also evaluated the cognitive demand of each test using the cognitive demand rating scale from the Surveys of Enacted Curriculum (SEC) framework (Porter, 2002). Here, we coded each test item based on its expectations for student performance, categorized using five levels of cognitive demand: (1) *memorize*, (2) *perform procedures*, (3) *communicate understanding*, (4) *solve non-routine problems*, and (5) *conjecture/generalize/prove*.[5] Raters assigned each item one score for this dimension, representing their judgment of the category that best matched the type of cognitive demand that the item posed to students. Each score point on the SEC scale was thus given a numeric value such that the scale is ordinal. Therefore, higher average SEC scores for a test might suggest a test with fewer 'memorize' items and more 'communicate understanding' items. We consider higher values to represent higher levels of item cognitive demand. The test coding using the SEC cognitive demand rating scale showed the same pattern of results as coding using the MQI *Task Cognitive Activation* scale, so we present only the MQI *Task Cognitive Activation* results in the paper.

Online Appendix A References

American Educational Research Association/American Psychological Association/ National Council on Measurement in Education. (1999). *Standards for educational and psychological testing*. Washington, DC: AERA.

Bridgeman, B. (1992). A comparison of quantitative questions in open-ended and multiple-choice formats. *Journal of Educational Measurement, 29*(3), 253-271.

Polikoff, M. S., Porter, A. C., and Smithson, J. (2011). How well aligned are state assessments of student achievement with state content standards? *American Educational Research Journal, 48*(4), 965-995.

Porter, A. C. (2002). Measuring the content of instruction: Uses in research and practice. *Educational Researcher, 31*(7), 3-14. doi: 10.3102/0013189X031007003

Webb, N. L. (1999). *Alignment of science and mathematics standards and assessments in four states* (Research Monograph No. 18). Madison, WI: National Institute for Science Education.

Webb, N. L. (2002). *Alignment study of language arts, mathematics, science, and social studies of state standards and assessments in four states*. Washington, DC: Council of Chief State School Officers.

Wixson, K. K., Fisk, M. C., Dutro, E., and McDaniel, J. (2002). *The alignment of state standards and assessments in elementary reading.* Ann Arbor, MI: Center for the Improvement of Early Reading Achievement.

Online Appendix B:
Results of Test Coding for Dimensions Included in the Summary *Overall Test Demand* Metric

Percentages of Items in Different Formats by Test and Grade.

	Multiple Choice			Short Answer			Open-Ended		
	Grade 4	Grade 5	Grades 4 and 5	Grade 4	Grade 5	Grades 4 and 5	Grade 4	Grade 5	Grades 4 and 5
State Tests									
Districts 1 and 2	62	64	63	19	12	16	19	24	22
District 3	86	86	86	12	12	12	2	2	2
District 4	100	100	100	0	0	0	0	0	0
District 5	100	100	100	0	0	0	0	0	0
Common Assessment									
All Districts	83	80	82	17	20	18	0	0	0

Figure B-1. Percentage of Items in Different Formats by Test and Grade.

Average Alignment to MQI Task Cognitive Activation by Test and Grade

	Grade 4	Grade 5	Grades 4 and 5
State Tests			
Districts 1 and 2	1.58	1.72	1.65
	0.76	*0.94*	*0.84*
District 3	1.16	1.25	1.21
	0.52	*0.63*	*0.57*
District 4	1.3	1.21	1.25
	0.57	*0.59*	*0.58*
District 5	1.18	1.3	1.24
	0.56	*0.58*	*0.57*
Common Assessment			
All Districts	1.3	1.46	1.38
	0.55	*0.69*	*0.63*

Note: Standard deviations in italics below averages.

Figure B-2. Average Alignment to MQI Task Cognitive Activation by Test and Grade.

Average SEC Cognitive Demand by Test and Grade

	Grade 4	Grade 5	Grades 4 and 5
State Tests			
Districts 1 and 2	2.38	2.36	2.37
	0.7	*0.86*	*0.77*
District 3	2.05	2.13	2.09
	0.6	*0.69*	*0.64*
District 4	2.25	2	2.11
	0.91	*0.66*	*0.78*
District 5	2.1	2.04	2.07
	0.71	*0.81*	*0.76*
Common Assessment			
All Districts	2.39	2.52	2.46
	0.68	*0.81*	*0.75*

Note: Standard deviations in italics below averages.

Figure B-3. Average SEC Cognitive Demand by Test and Grade.